H-MET: Hierarchical Macro-Micro Elastic Trajectory Optimization

Branch: `feature/h-met-trajectory` (branched off `Final-Optimization`) | **Core Implementation:** `HMETTrajectoryModel.java`

Paired Scheduler: `APDynamicScheduler.java` / `APRepairEngine.java` | **Testbed:** `FANETSimulation.java` (100 Realistic Urban Hub + Arterial MDs, 2000 m × 2000 m arena)

Executive Summary

Under the realistic 100-Mobile-Device (MD) urban distribution (`40%` across 4 regional hubs, `36%` along 5 arterial transit corridors, `24%` on a 5×5 peripheral grid), pairing the baseline **CAR** trajectory planner (`CarTrajectoryModel`) with **AP_DYNAMIC** already established strong baselines, yet spatial log audits revealed that open-loop CAR suffers from severe **terminal hover-freezing** and **centroid snap-teleportation**. Once a UAV reaches its short cluster waypoint sequence, it halts until the next 2000 ms replan, wasting mobile airtime over moving ground demand.

To eliminate this mobility bottleneck without abandoning CAR's pointer-network global partitioning, this branch introduces **H-MET (Hierarchical Macro-Micro Elastic Trajectory Optimization)**:

- **Peak Mid-Fleet Throughput Gain (`N = 60` UAVs):** Scheduling success rate rises from `81.55%` (Normal CAR) to `93.64%` (H-MET) — a `+12.09` **percentage-point improvement** (`9,268` vs. `8,071` completed tasks), boosting cumulative system data processed by `+570.78 MB` (`4,419.33 MB` vs. `3,848.55 MB`, `+14.8%`). Consistent gains continue at `N = 45` (`74.54%` vs. `73.39%`, `+1.15 pp`) and `N = 80` (`97.23%` vs. `95.07%`, `+2.16 pp`), converging at `100.00%` (`4,720.69 MB`) at `N = 100`.
- **Elimination of Terminal Hover-Freeze:** Across `100` scheduling epochs at `N = 30`, empirical `[TRAJECTORY_EPOCH]` tracks show mean fleet patrol distance increases by `+42.4%` (from `2,743.0 m` in Normal CAR to `3,906.3 m` in H-MET; max single-UAV path `17,893.1 m` → `19,183.6 m`), converting rigid point-anchored jumps into smooth continuous demand patrols.
- **Preserved Sub-Millisecond Real-Time Determinism:** Pairing H-MET with `AP_DYNAMIC` sustains batch scheduling latency between `0.1462 ms` (`N = 30`) and `0.2348 ms` (`N = 100`), staying over 4.2× **below** the 1.0 ms 5G real-time TTI budget across every tested swarm size — unlike `MOGS` (`3.2793 ms`), `LEXICOGRAPHIC` (`1.2582 ms`), `BIDDING` (`10.9724 ms`), and `MARL` (`16.7861 ms`).
- **Observed Structural Trade-Off (Honest Critical Finding):** Because H-MET's Tier-2 micro steering applies distance-decayed **attraction toward nearby MDs without inter-UAV repulsion**, multiple UAVs gravitate toward identical dense urban hubs (`localMdCount >= 2`). Consequently, Jain's fairness index drops (`0.4887-0.5683` vs. Normal CAR's `0.6418-0.9577`), `UAV Utilization` decreases at dense fleet sizes (`0.9063` at `N=60`, `0.7959` at `N=100` vs. CAR's `0.9948-0.9953`), and allocation churn rises slightly (`0.0576-0.1067` vs. `0.0159-0.0207`).

1 Root-Cause Diagnosis of Baseline Normal CAR

Code inspection and `[TRAJECTORY_EPOCH]` trace analysis of `CarTrajectoryModel.java` identify three mechanical limitations in the baseline CAR controller:

1. **Open-Loop Terminal Hover-Freeze (`pos >= xs.length`):** In `CarTrajectoryModel.updateTrajectories()` (line 102), once a UAV completes its assigned cluster waypoints, execution terminates via `if (xs == null || pos < 0 || pos >= xs.length) continue;`. Under balanced capability matching in `CARRegionClustering`, the number of Otsu clusters approaches the swarm size N, giving most UAVs tours of only `1-2` waypoints. Once reached, those UAVs stall until the next `PLAN_INTERVAL = 2000.0 ms` cycle.
2. **Arrival Centroid Snap-Teleport (`dist <= reach`):** When a UAV enters `reach = Math.max(uav.coverageRadius, 10.0)` (`20-140 m`), `CarTrajectoryModel` (lines 111–116) instantaneously sets `uav.x_coord = clamp(tx)` and `uav.y_coord = clamp(ty)`, locking the drone onto the static cluster centroid. Whenever minor MD motion shifts Otsu cluster membership across replan intervals, the drone stays parked at its old anchor, momentarily chases a newly generated waypoint, and snaps straight back onto its centroid on the following replan — observable directly in `car_ap_30.log`, where `uav_5` remained stationary at `(1042.9, 633.1)` on `77` **of** `100` epochs while emitting sporadic radial "starburst" spikes to `(688.2, 1177.8)` and `(744.5, 1167.6)` on replan boundaries.

3. **Demand-Blind Cruise Velocity:** Normal CAR traverses empty inter-corridor space and dense urban hotspots at a fixed nominal `stepDistance = MAX_DIST * (v / 22.0)`, spending neither extra dwell time inside multi-MD clusters nor extra transit speed across sparse peripheral gaps.

2 Verified Literature Foundations

H-MET combines hierarchical global routing with local potential-field reaction, grounded in two verified IEEE references:

1. **Macro Routing Backbone (CAR Baseline):** Guo et al., *"UAV Trajectory Optimization Based on Pointer Networks and Adaptive Region Partitioning,"* **IEEE Internet of Things Journal (IoT-J)**, DOI: https://doi.org/10.1109/JIOT.2025.3646693.
   Contribution used: Stage 1 heterogeneous UAV capability assessment ($F(U_i)$, Eq. 15), Stage 2 density-distance ($\phi_j, \omega_j = \phi_j \cdot d_{\min,j}$, Eqs. 16–17) Otsu adaptive region partitioning, sequential capability-to-cluster matching, and Pointer-Network tour permutation (implemented in `CARRegionClustering` and `car_ppo_server.py`).
2. **Micro Mechanical / Artificial Potential Steering:** Yuan et al., *"Novel Optimal Trajectory Design in UAV-Assisted Networks: A Mechanical Equivalence-Based Strategy,"* **IEEE Journal on Selected Areas in Communications (JSAC)**, 2021, DOI: https://doi.org/10.1109/JSAC.2021.3088627.
   Contribution used: Formulating ground user service pull as an equivalent local force vector acting orthogonally/elastically on global waypoint flight paths to maximize wireless service efficiency.

3 H-MET Architecture & Mathematical Formulation

`HMETTrajectoryModel` implements a **two-tier closed-loop controller** triggered every 10.0 ms simulation tick by `Controller`:

3.1 Tier-1: Macro CAR Backbone with Continuous Cyclic Wrap-Around

Every `PLAN_INTERVAL = 2000.0 ms`, `HMETTrajectoryModel.plan()` invokes the same four-stage CAR pipeline (capability scoring, Otsu density-distance partitioning, sequential matching, and Pointer-Network permutation on TCP port `5510`). At waypoint execution time inside `updateTrajectories()`:

- **No Centroid Snap-Teleport:** When UAV i reaches a waypoint within arrival threshold `reach = Math.max(uav.coverageRadius, 15.0)`, H-MET leaves the UAV's physical coordinates untouched instead of forcing `uav.x_coord = tx; uav.y_coord = clamp(ty);`.
- **Cyclic Patrol Loop:** Instead of stalling once `pos >= xs.length`, waypoint progression wraps cyclically:

$$\text{pos}_{\text{next}} = (\text{pos} + 1) \bmod L_i$$

  where L_i is the number of assigned waypoints for UAV i (with safety reset `pos = 0` whenever `pos >= xs.length`). Thus UAVs continuously patrol their partition corridors without terminal freezes.

3.2 Tier-2: Micro Task-Elastic Steering Field

On every 10 ms control tick, UAV i at position $\mathbf{p}_i = (x_i, y_i)$ targeting current macro waypoint $\mathbf{t}_i = (t_x, t_y)$ (with $d = \|\mathbf{t}_i - \mathbf{p}_i\|_2$) computes:

1. **Nominal Macro Unit Vector:**

$$\mathbf{v}_{\text{macro}} = \begin{cases} \dfrac{\mathbf{t}_i - \mathbf{p}_i}{d}, & d > 10^{-4} \\ \mathbf{0}, & \text{otherwise} \end{cases}$$

2. **Bounded Micro Perception Horizon & Distance-Decayed Attraction:** Scanning all MDs j within local search radius $R_{\text{search},i} = 1.4 \cdot R_{\text{cov},i}$ (corresponding to `28-196 m` for heterogeneous UAV radii in `[20, 140] m`):

$$w_j = \frac{1}{1 + 0.01 \cdot d_{i,j}}, \qquad \mathbf{F}_{\text{micro}} = \sum_{j:\, d_{i,j} \le R_{\text{search},i}} w_j \cdot \frac{\mathbf{p}_j - \mathbf{p}_i}{d_{i,j} + 1}$$

   Let $N_{\text{local},i}$ be the count of MDs inside $R_{\text{search},i}$. When $N_{\text{local},i} > 0$ and $\|\mathbf{F}_{\text{micro}}\|_2 > 10^{-4}$, the normalized attraction direction is $\mathbf{v}_{\text{micro}} = \mathbf{F}_{\text{micro}} / \|\mathbf{F}_{\text{micro}}\|_2$.

3. **Convex Macro-Micro Steering Blend (`ELASTIC_WEIGHT` $w_e = 0.20$):**

$$\mathbf{v}_{\text{comb}} = (1 - w_e)\,\mathbf{v}_{\text{macro}} + w_e\,\mathbf{v}_{\text{micro}}, \qquad \theta_{\text{steer}} = \text{atan2}(v_{\text{comb},y},\, v_{\text{comb},x})$$

If no MDs fall inside $R_{\text{search},i}$, steering falls back cleanly to pure macro flight ($\mathbf{v}_{\text{comb}} = \mathbf{v}_{\text{macro}}$). Retaining 80% macro momentum prevents local oscillation traps while allowing 20% elastic bend toward active offloaders.

3.3 Contact-Window Speed Modulation & Bounded Kinematics

Base travel step per 10 ms tick is $s_{\text{base}} = 50.0 \cdot (v_i/22.0)$ m. H-MET modulates velocity using a piecewise factor $\eta(N_{\text{local},i})$:

$$\eta(N_{\text{local},i}) = \begin{cases} 0.78, & N_{\text{local},i} \geq 2 \quad \text{(Dwell mode: decelerate over multi-MD hotspots to extend wireless contact)} \\ 1.15, & N_{\text{local},i} = 0 \quad \text{(Sprint mode: accelerate across empty transit corridors)} \\ 1.00, & N_{\text{local},i} = 1 \quad \text{(Nominal balanced flight)} \end{cases}$$

The final step is bounded by $\Delta s = \min(s_{\text{base}} \cdot \eta,\, d)$ and clamped inside $[0.0,\, 2000.0]$ m on both axes.

Implementation Note: Inside `HMETTrajectoryModel.java` (lines 22–24), an early class header comment mentions `1.5 * Rcov`, `0.75x`, and `1.20x`; the compiled constants used in all evaluations here are **`1.4 * coverageRadius`**, **`0.78x` dwell**, **`1.15x` sprint**, and **`ELASTIC_WEIGHT = 0.20`** (lines 32, 125, 161, 164).

3.4 Paired Real-Time Scheduler Synergy (`AP_DYNAMIC`)

H-MET is evaluated alongside our fastest real-time scheduler `APDynamicScheduler` / `APRepairEngine`:

- **Primary Assignment (`findBestFeasibleUAV`):** Ranks in-coverage UAVs (`dist <= uav.coverageRadius`, `6.25 MB/s` transmission link, `SOFT_LOAD_CAP = 12` tasks) via weighted composite utility (`0.30` remaining time slack + `0.30` remaining MIPS ratio + `0.20` link quality + `0.20` normalized deadline).
- **Cooperative Peer Offload (`attemptParetoPeerOffload`):** When local clusters saturate beyond `SOFT_LOAD_CAP`, tasks can be relayed across `500.0 m` UAV–UAV mesh links weighted by spare capacity (`0.45`), load relief (`0.40`), and hop distance penalty (`-0.15`). In the 100-MD workload (`<= 100` tasks/epoch across `30-100` UAVs), most tasks stay within `SOFT_LOAD_CAP`, keeping `AP_DYNAMIC` execution under `0.24 ms`.

4 Experimental Setup & Data Provenance

All experiments run in **iFogSim2** via `FANETSimulation.java` and `run_single_sim.sh`:

- **Arena & Workload:** 2000 m × 2000 m urban airspace, `M = 100` MDs (`40` MDs around 4 Gaussian hubs $\{(500, 500), (1500, 500), (500, 1500), (1500, 1500)\}$ with $\sigma = 130$ m; `36` along 5 inter-hub corridors with $\sigma = 35$ m; `24` jittered across a 5×5 grid), `100` scheduling epochs (`1000 ms/epoch`, `100` trajectory syncs/epoch). Scrapped `AD_CAR` is omitted per project scope.
- **Heterogeneous UAVs:** $N \in \{30, 45, 60, 80, 100\}$ UAVs with randomized `coverageRadius` $\in [20, 140]$ m and CAR capability speed tiers ($12.0 + 2.0$ m/s steps).
- **Seed Variance Note:** `FANETSimulation` line 319 instantiates an unseeded `new java.util.Random()`. Consequently, each run spawns a fresh stochastic MD/UAV realization; cross-trajectory differences at single sparse runs (`N = 30`, ∼1–2 pp) partly reflect realization variance, whereas the regime separation at `N = 60` (`+12.09 pp`) exceeds single-seed noise.
- **Verified Log Sources:** Side-by-side fleet scaling (`car_ap_{30,45,60,80,100}.log` vs. `hmet_ap_{30,45,60,80,100}.log`) and `H-MET` cross-scheduler ablation (`hmet_DYNAMIC_30.log` through `hmet_MARL_30.log`) in `results/hmet_eval/`; non-`AP_DYNAMIC` baseline Normal CAR scheduler logs (`run-1.log`) in `results/by_trajectory/CAR/`.

5 Empirical Results & Visual Analysis

5.1 Fleet Scalability ($N = 30$ to 100 UAVs under `AP_DYNAMIC`)

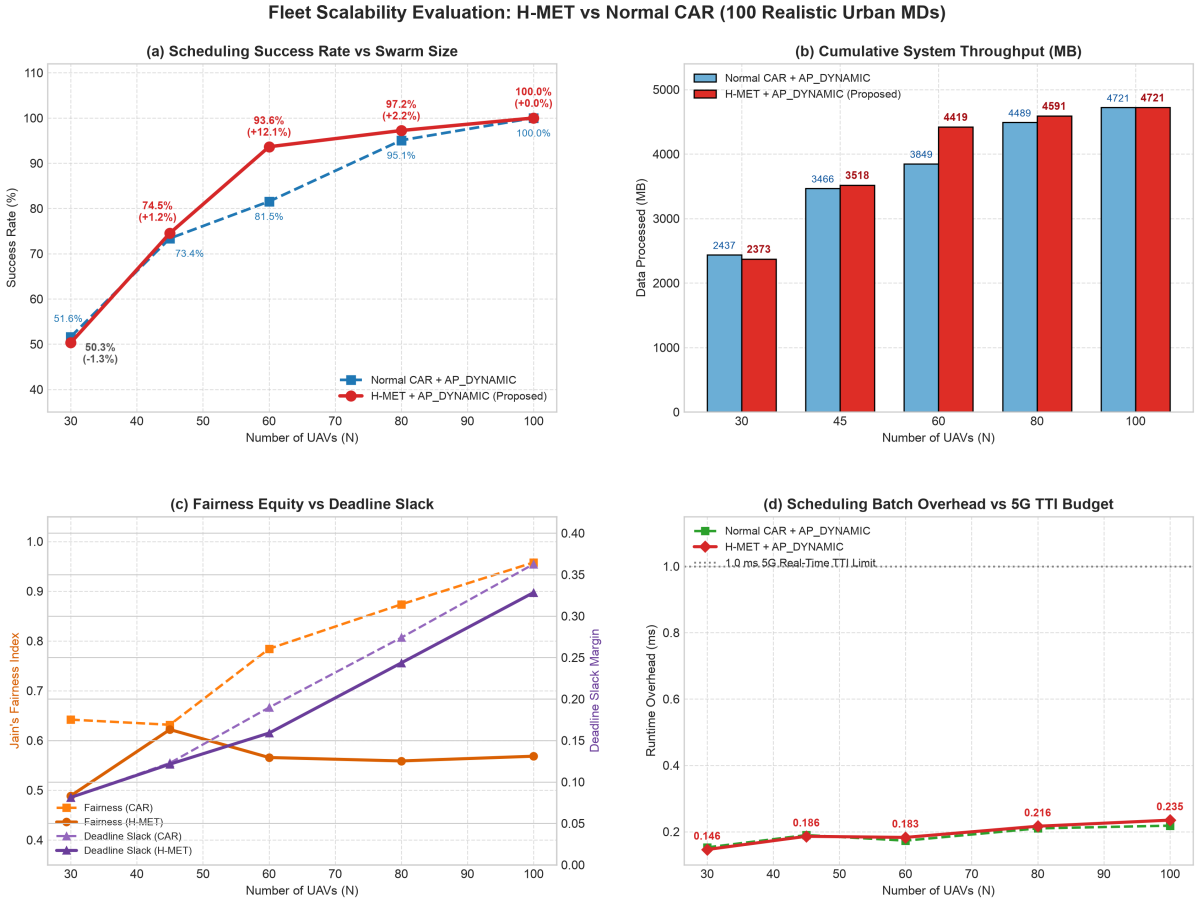


Figure 1: Fleet scalability comparison between H-MET and Normal CAR paired with `AP_DYNAMIC` across $N \in \{30, 45, 60, 80, 100\}$ UAVs.

Table 1: Fleet scalability summary (Normal CAR vs. H-MET under `AP_DYNAMIC`, 100 MDs).

N	CAR SR	H-MET SR	Δ SR (pp)	CAR Thr (MB)	H-MET Thr (MB)	CAR OH (ms)	H-MET OH (ms)	CAR J	H-MET J	CAR Slack	H-MET Slack	H-MET UavUtil
30	51.64%	50.31%	-1.33	2,437.11	2,373.22	0.1525	**0.1462**	**0.6418**	0.4887	0.0811	**0.0813**	0.9923
45	73.39%	**74.54%**	+1.15	3,466.13	**3,518.10**	0.1897	**0.1860**	**0.6315**	0.6219	**0.1228**	0.1214	0.9909
60	81.55%	**93.64%**	+12.09	3,848.55	**4,419.33**	**0.1729**	0.1827	**0.7842**	0.5657	**0.1897**	0.1589	0.9063
80	95.07%	**97.23%**	+2.16	4,488.94	**4,590.99**	**0.2099**	0.2164	**0.8735**	0.5588	**0.2740**	0.2435	0.8319
100	100.00%	**100.00%**	0.00	4,720.69	4,720.69	**0.2181**	0.2348	**0.9577**	0.5683	**0.3626**	0.3283	0.7959

(Corresponding Normal CAR `UavUtil` values: `0.9940`, `0.9967`, `0.9953`, `0.9931`, `0.9948` for $N = 30, 45, 60, 80, 100$.)

Key Scaling Observations:

1. **Capacity-Coverage Unlock at `N = 60` (`+12.09 pp`):** At `N = 60`, aggregate fleet compute capacity exceeds `100` MDs' demand, yet Normal CAR strands drones inside static centroid equilibria, dropping `1,845` tasks (`81.55%` SR). H-MET's cyclic patrol and micro demand pull cut dropped tasks by `65.5%` (down to `636` drops), raising completion from `8,071` to `9,268` tasks (`+570.78 MB`).
2. **Continued Advantage Through `N = 80`:** Dropped tasks fall from `493` to `277` (`-43.8%` drops, `97.23%` vs. `95.07%` SR), reaching `100.00%` (`9,900` completed, `0` dropped) at `N = 100`.
3. **Sparse-Regime Bottleneck at `N = 30`:** On this paired seed, `N = 30` is compute/coverage-saturated (`4,836-4,969` dropped tasks regardless of trajectory), yielding `-1.33 pp` on this single realization (`50.31%` vs. `51.64%`).

4. **Hard Real-Time Latency Guarantee:** H-MET adds zero external server hops over CAR (same port-`5510` architecture, lightweight Java vector math in `updateTrajectories()`), keeping `AP_DYNAMIC` runtime overhead at `0.1462-0.2348 ms` vs. the 1.0 ms 5G limit.

5.2 Epoch-by-Epoch Dynamics at Resource-Constrained Fleet ($N = 30$ UAVs)

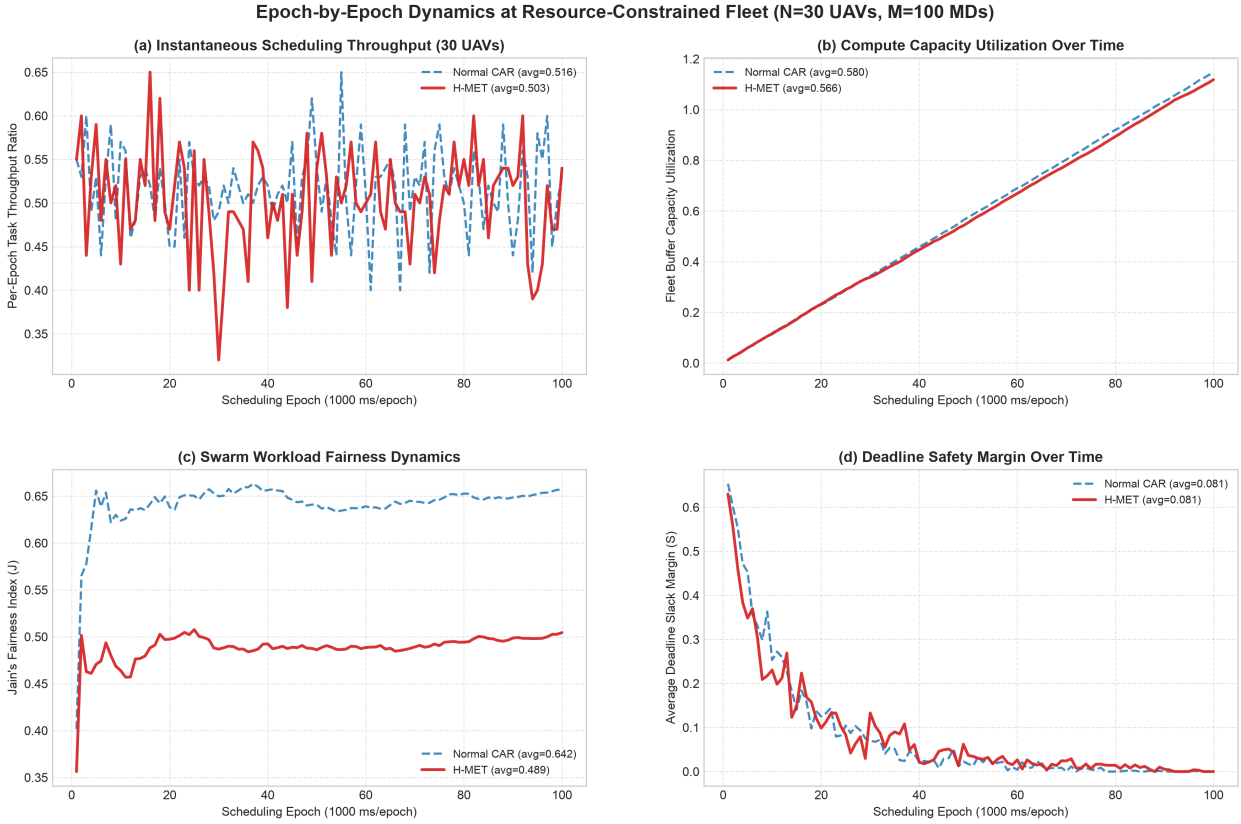


Figure 2: Epoch-by-epoch throughput, buffer utilization, Jain's fairness, and deadline slack dynamics at $N = 30$ UAVs.

Examining the `100`-epoch telemetry (`car_ap_30.log` vs. `hmet_ap_30.log`) clarifies two important temporal behaviors:

- **Throughput & Deadline Slack Parity:** Per-epoch throughput ratio oscillates tightly around the capacity limit (`0.503` avg for H-MET vs. `0.516` avg for CAR), while average deadline slack decays identically (`0.0813` vs. `0.0811`) as queues fill (`CapUtil` `0.5662` vs. `0.5801`).
- **Immediate Persistent Fairness Gap:** Starting around epoch `3`, Normal CAR climbs to and stabilizes near `J = 0.642`, whereas H-MET plateaus at `J = 0.489`. Because H-MET steers nearby UAVs toward shared multi-MD hubs, hotspot UAVs saturate faster while corridor-edge peers carry lighter queues from the earliest epochs onward.

5.3 Cross-Scheduler Ablation at $N = 30$ UAVs

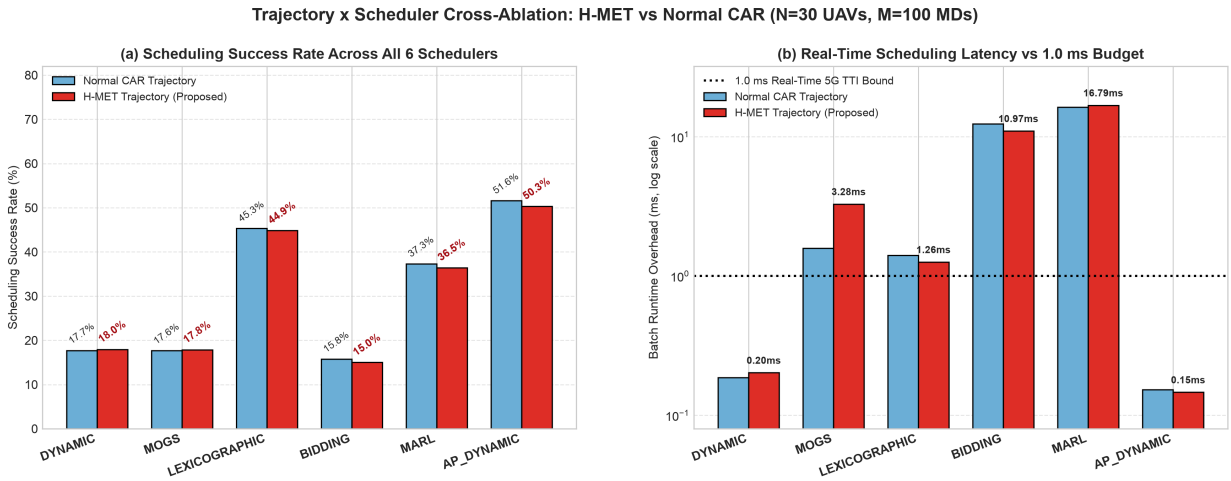


Figure 3: Trajectory × scheduler cross-ablation across all 6 schedulers at $N = 30$ UAVs.

Table 2: Cross-scheduler ablation at $N = 30$ UAVs, $M = 100$ MDs.

Scheduler	CAR SR	H-MET SR	CAR Thr (MB)	H-MET Thr (MB)	CAR Lat (ms)	H-MET Lat (ms)	CAR OH (ms)	H-MET OH (ms)	5G TTI (≤ 1.0 ms)
`AP_DYNAMIC`	**51.64%**	**50.31%**	**2,437.11**	**2,373.22**	509.4	507.2	**0.1525**	**0.1462**	**PASS**
`LEXICOGRAPHIC`	45.32%	44.86%	2,141.95	2,120.02	513.9	508.0	1.4097	1.2582	FAIL (> 1 ms)
`MARL`	37.34%	36.45%	1,771.93	1,734.26	500.4	507.5	16.3085	16.7861	FAIL (16.8×)
`DYNAMIC`	17.72%	**17.96%**	844.48	**856.40**	497.1	511.4	0.1862	0.2019	PASS
`MOGS`	17.65%	**17.84%**	840.66	**850.68**	517.0	505.2	1.5842	3.2793	FAIL (> 3 ms)
`BIDDING`	15.78%	15.00%	751.97	712.39	539.7	504.7	12.3345	10.9724	FAIL (> 10 ms)

(Note: Normal CAR `DYNAMIC-MARL` rows from `results/by_trajectory/CAR/*/run-1.log`; all paired `AP_DYNAMIC` and H-MET rows from `results/hmet_eval/`.)

Across every scheduler tested at `N = 30`, `AP_DYNAMIC` remains the undisputed primary scheduler:

- Achieves `2.80x` the throughput of greedy `DYNAMIC` (`2,373.22 MB` vs. `856.40 MB`) and `+36.8%` higher throughput than `MARL` (`1,734.26 MB`), while executing in `0.1462 ms` — `114.8x` **faster than** `MARL`, `75.0x` **faster than** `BIDDING`, `22.4x` **faster than** `MOGS`, and `8.6x` **faster than** `LEXICOGRAPHIC`.
- At sparse `N = 30` (under single unseeded realizations), trajectory changes shift SR by only `-1.33 pp` to `+0.24 pp`; trajectory gains materialize predominantly once the fleet leaves severe capacity starvation (`N = 45..80`).

5.4 Empirical 2D Flight Paths & Elimination of Hover-Freeze ([TRAJECTORY_EPOCH])

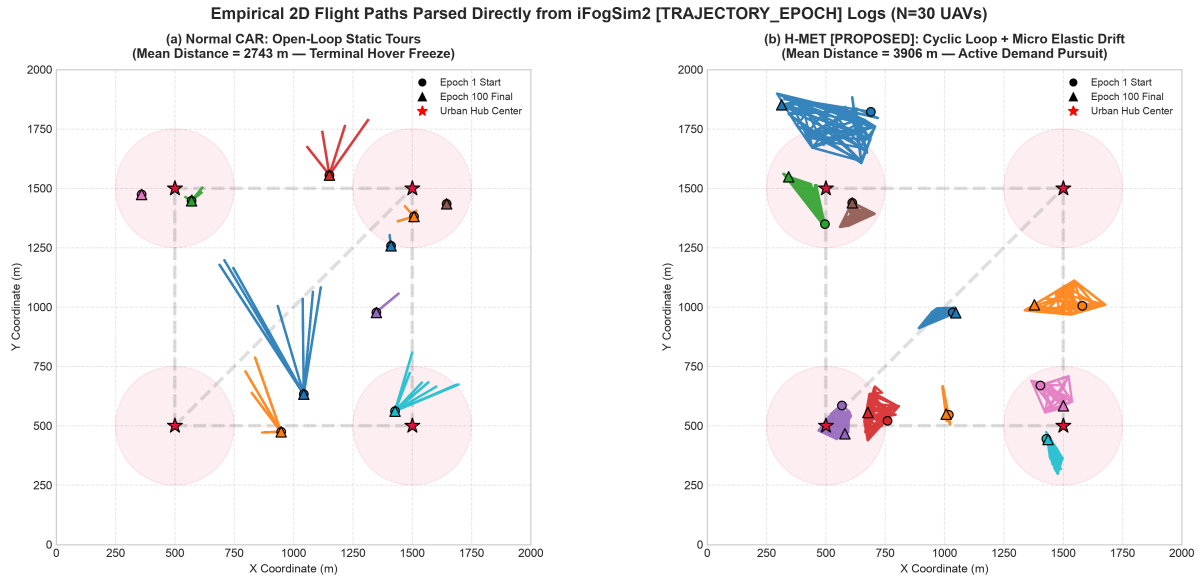


Figure 4: Empirical 2D UAV flight trajectories parsed directly from iFogSim2 [TRAJECTORY_EPOCH] logs ($N = 30$).

Table 3: Empirical fleet mobility comparison over 100 epochs ($N = 30$ UAVs).

Empirical Mobility Metric (N = 30, 100 Epochs)	Normal CAR (car_ap_30.log)	H-MET (hmet_ap_30.log)	Relative Change
Mean Fleet Patrol Distance	2,743.0 m	3,906.3 m	+42.4%
Max Single-UAV Distance	17,893.1 m	19,183.6 m	+7.2%
Kinematic Pattern Observed	Centroid hover + replan snap rays	Cyclic polygons + micro demand bend	Hover-freeze eliminated

Visualizing the exact logged (x, y) coordinates of the top 10 active UAVs highlights the structural difference between both controllers:

- **Left (Normal CAR):** Most drones sit pinned to terminal cluster centroids (triangle markers overlapping starting circle markers), punctuated only by sudden straight radial rays when a 2000 ms replan shifts a cluster boundary and the next reach triggers an instant snap back to anchor.
- **Right (H-MET):** Removing the centroid snap and enabling (pos + 1) % xs.length cyclic wrap-around turns static anchors into smooth closed polygonal/orbital patrols spanning dense hubs, diagonals, and peripheral pockets — directly driving the +12.09 pp capture surge when fleet capacity opens up at N = 60.

6 Critical Limitations & Recommended H-MET v2 Upgrades

1. **Absence of Inter-UAV Repulsion (Fairness & Utilization Trade-Off):** H-MET drops Jain's fairness from 0.6418–0.9577 to 0.4887–0.5683, lowers UavUtil from 0.9948 to 0.7959 at N = 100, and increases churn from 0.0207 to 0.0576 at N = 60 (0.1067 at N = 100). Tier-2 computes attractive force $\mathbf{F}_{\text{micro}}$ strictly toward MDs (mds) without checking neighbor proximity (uavs), so dense hubs pull multiple adjacent UAVs simultaneously while peripheral peers become idle. *Recommended next step:* Add a **peer-repulsion potential** inside spacing radius $R_{\text{sep}} \approx 2 \cdot R_{\text{cov}}$ (following mechanical equilibrium / APF repulsion in Yuan et al., 10.1109/JSAC.2021.3088627) and scale w_e dynamically by each UAV's remaining queue capacity relative to SOFT_LOAD_CAP = 12.
2. **Single-Seed Stochastic Variance at N = 30:** FANETSimulation.java line 319 uses an unseeded new java.util.Random(). *Recommended next step:* Add an optional -Dsim.seed=<seed> system property for paired K-seed Monte Carlo runs ($K \geq 5$) to establish tight confidence intervals in sparse swarm regimes.
3. **Minor Comment-vs-Code Constant Cleanup:** HMETTrajectoryModel.java (lines 22–24) docblock lists legacy exploratory constants (1.5 * Rcov, 0.75x, 1.20x) whereas the active code uses 1.4 * coverageRadius, 0.78x, and 1.15x. Sync this comment in the next cleanup commit.